



CSE-476: Introduction to Natural Language Processing
Quiz 3

Multiple Choice Questions (Single correct)

Instructions: Select the correct option with a short explanation.

- Q1. In a standard RNN, what is the primary purpose of the "hidden state"?
- A. To store the final output of the entire sequence.
 - B. To act as a memory that carries information from previous time steps
 - C. To calculate the loss function for the current word
 - D. To normalize the input embeddings
- Q2. Why is the Softmax function typically used in the final layer of a Neural Language Model?
- A. To ensure the output values sum to 1, representing a probability distribution
 - B. To prevent the gradients from exploding
 - C. To increase the speed of the training process
 - D. To allow the model to handle variable-length sequences
- Q3. What is a key limitation of feed-forward neural networks for NLP tasks?
- A. They cannot use embeddings
 - B. They assume fixed input size and struggle with temporal order
 - C. They cannot be trained with gradient descent
 - D. They do not use activation functions

- Q4. What is the purpose of the "Forward Pass" in training?
- A. To update weights based on the loss
 - B. To calculate the predicted output and compute the loss
 - C. To unroll the network into a sequence
 - D. To calculate the gradient for each parameter
- Q5. What is true about the weights U , V , and W across different timestamps in an RNN?
- A. They are randomly re-initialized at every step
 - B. They are reused across timestamps
 - C. They increase in value as the sequence progresses
 - D. Only the output weight V is reused
- Q6. Suppose all activation functions in a 3-layer neural network are linear. What is the effective model equivalent to?
- A. A deeper network with more representational power
 - B. A single linear model
 - C. A recurrent neural network
 - D. A decision tree
- Q7. When training an RNN for language modeling using teacher forcing, the model predicts:
- A. The next word given the predicted history
 - B. The next word given the ground-truth history
 - C. The current word given itself
 - D. All words simultaneously
- Q8. In an RNN, the hidden state at time t depends on:
- A. Only the input at time t
 - B. Only the hidden state at time $t - 1$
 - C. Both the input at time t and the hidden state at time $t - 1$
 - D. All future inputs

- Q9. Consider a neural network layer defined as: $h = f(Wx)$. The input vector x has shape $(n_0 \times 1)$, and the output vector h has shape $(n_1 \times 1)$. What must be the shape of W ?
- A. $(n_0 \times n_1)$
 - B. $(n_1 \times n_0)$
 - C. $(n_0 \times 1)$
 - D. $(1 \times n_1)$
- Q10. What is the main role of backpropagation in neural network training?
- A. To compute predictions during inference
 - B. To calculate gradients of the loss with respect to parameters
 - C. To initialize weights randomly
 - D. To normalize input data